



2020 QATS Maths Advanced HSC Trial Paper

Mathematics Advanced (Willoughby Girls High School)



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Student Name:.....

2020 Higher School Certificate Trial Examination Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11-36, show relevant mathematical reasoning and/or calculations

**Total marks:
100**

Section I 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II 90 marks

- Attempt Questions 11-36
- Allow about 2 hours and 45 minutes for this section

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MATR20_EXAM

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet provided for Questions 1-10

1 Differentiate $x^4 + e^x$

A. $4x^3 + \ln x$

B. $4x^3 + e^x$

C. $4x^3 e^x$

D. $4x^3 + xe^x$

2 Simplify $\frac{x^2 + 5x + 6}{x^2 - 9}$

A. $\frac{x+2}{x-3}$

B. $\frac{x+3}{x-3}$

C. $\frac{x+2}{x+3}$

D. $\frac{x-2}{x-3}$

3 The amplitude of the curve $y = \frac{1}{2} \sin\left(3x + \frac{\pi}{6}\right)$ is

A. $\frac{1}{6}$

B. $\frac{\pi}{6}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

4 The mean of a set of data is 14 and the standard deviation is 2.1.

If each score in the data set is increased by 4, which of the following statements will be true?

A. The mean and standard deviation will increase by 4

B. The mean will increase by 4 and the standard deviation will not change

C. The mean will not change, and the standard deviation will increase by 4

D. The mean and standard deviation will increase by a factor of 4

5 A discrete probability distribution is shown in the table below.

$X = x$	1	2	3	4	5
$P(X = x)$	0.23	0.38	0.21	0.13	0.05

The expected value is

A. 1.00

B. 1.56

C. 2.39

D. 5.71

6 Evaluate $\int_0^2 (3x^4 + 2x)dx$

A. $\frac{96}{5}$

B. $\frac{116}{5}$

C. $\frac{136}{5}$

D. 100

7 If $\sin x = \frac{5}{13}$ and x is an acute angle, then $\cos x$ is equal to

A. $\frac{12}{13}$

B. $\frac{13}{5}$

C. $\frac{5}{12}$

D. $\frac{12}{5}$

8 Given $\log_3 5 = 1.46$ and $\log_3 2 = 0.63$, evaluate $\log_3 100$

A. 2.09

B. 2.72

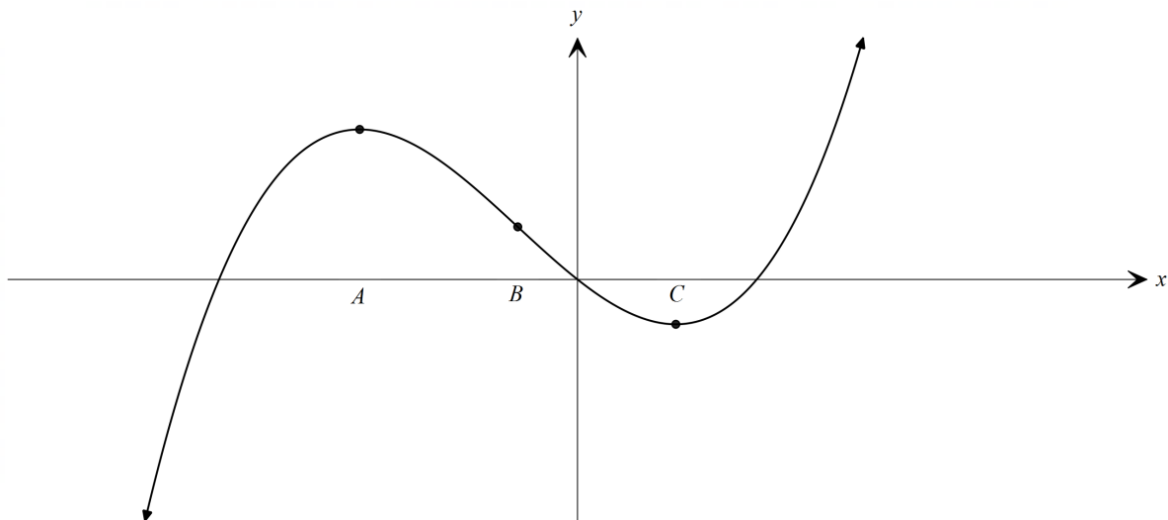
C. 3.55

D. 4.18

- 9 The graph of $y = f(x)$ undergoes a vertical dilation of a units, followed by a horizontal translation of b units to the right, and is then reflected in the x axis.

The resulting graph is

- A. $y = af(x+b)$
B. $y = -af(x-b)$
C. $y = -f(ax+b)$
D. $y = f(ax-b)$
- 10 The graph of $y = f(x)$ is shown below.
 $x = A$ and $x = C$ are stationary points, and $x = B$ is a point of inflection.



Over what domain is $f'(x) < 0$ and $f''(x) > 0$?

- A. $(-\infty, A)$
B. (A, B)
C. (B, C)
D. (C, ∞)

Section II

90 marks

Attempt Questions 11-36

Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

In Questions 11-36, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Find the derivative of $y = x^2 \sin x$

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Question 12 (2 marks)

Find $\int \cos 4x \, dx$

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Question 13 (3 marks)

The formula for first principles differentiation is

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$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Use this formula to find the derivative of $f(x) = x^2 + 3x$.

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Question 14 (2 marks)

The goals scored by a netball team each game is presented in the stem-and-leaf plot below.

Stem	Leaf
3	0 0 1 3
4	4 4 7 8 9
5	3 5

(a) Find the range for this set of data. 1

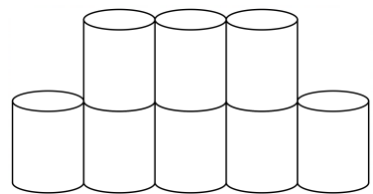
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(b) Find the median number of goals scored. 1

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Question 15 (3 marks)

A shop assistant is building a display of soup cans.



The top row will contain 3 cans, and each subsequent row below will have 2 more cans than the previous row above.

(a) How many cans would be in the 10th row? 1

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Question 15 continues on page 9

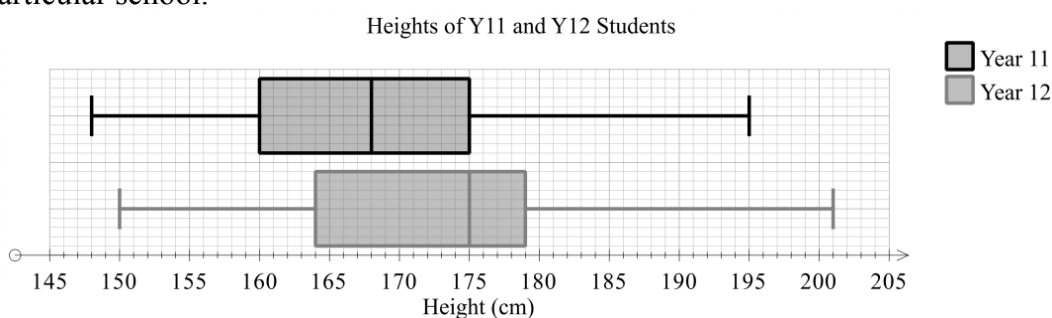
Question 15 (continued)

- 2

[illegible]

Question 16 (2 marks)

The boxplot below shows the heights of a students in Year 11 and Year 12 at a particular school.



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There are the same number of students above 175 cm in Year 11 as there are students above 175 cm in Year 12. If there are 140 students in total in Year 11, how many students are in Year 12?

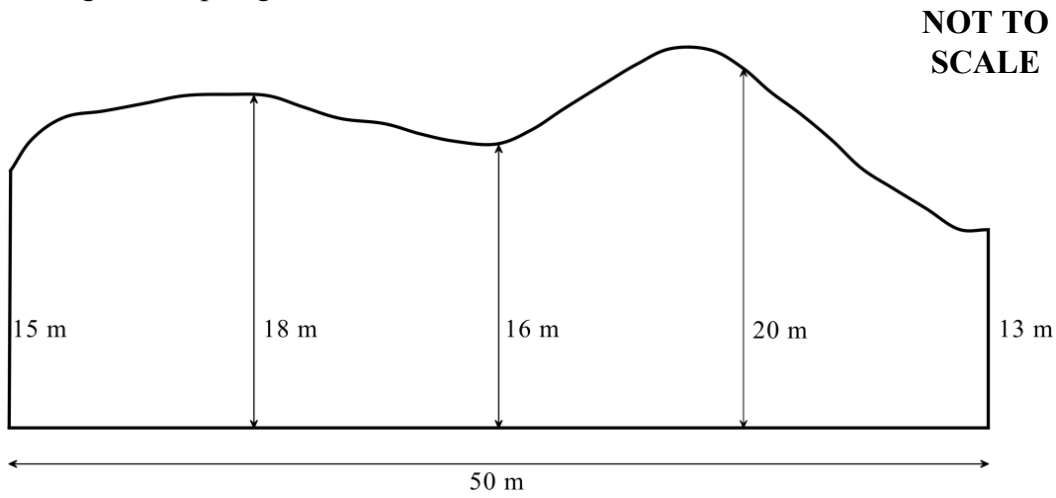
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Question 17 (3 marks)

An irregular shaped garden is shown below.



Use two applications of the trapezoidal rule to find an approximation for the area of the garden.

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Towns A , B and C are to be connected by high-speed optic fibre cables.
Town B is due west of town A .
Town C is on a bearing of 210° from Town A .



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Question 19 (7 marks)

A graph is defined by $y = \frac{x}{e^x}$.

- (a) Find any stationary points and determine their nature.

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Question 19 continues on page 13

Question 19 (continued)

- (b) Hence, sketch the graph of $y = \frac{x}{e^x}$, showing any stationary points, inflections, intercepts and asymptotes.

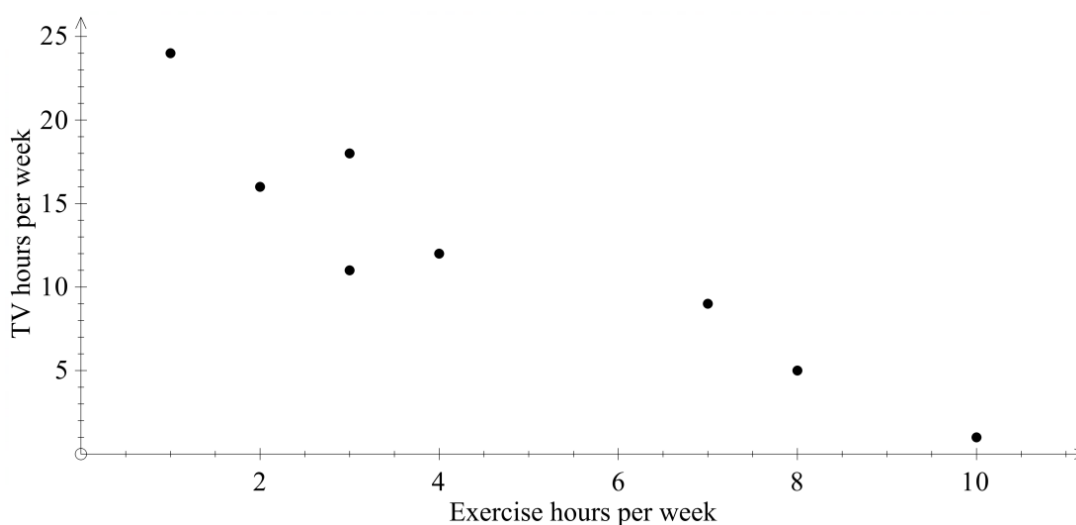
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[illegible]

Question 20 (2 marks)

A group of children were surveyed on the number of hours each week that they spent exercising and watching television. The results are shown in the table and scatterplot below.

Exercise hours	4	1	8	7	10	3	3	2
TV hours	12	24	5	9	1	18	11	16



- (a) Determine the equation of the least-squares regression line for this data. 1

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- (b) Using your equation from part (a), predict how many hours a week a child would spend exercising, correct to 1 decimal place, if they spend 15 hours watching television. 1

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Question 21 (3 marks)

The heights of a year group of 200 students are normally distributed with a mean of 162 cm and a standard deviation of 4 cm.

- (a) What percentage of students are taller than 170 cm? 1

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- (b) How many students are between 158 cm and 170 cm? **2**

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Question 22 (3 marks)

Solve $2\sin(3\theta) = -1$ for $0 \leq \theta \leq \pi$ **3**

[illegible]

Question 23 (5 marks)

The population of an ant colony is modelled by the equation $P = 1000e^{kt}$, where t is measured in weeks.

- (a) Find the initial population. **1**

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- (b) After 5 weeks the population is 15 000.
Show that $k = 0.5416$, correct to 4 decimal places. **2**

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- (c) During which week will the population exceed 50 000 for the first time? **2**

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Find $\int \frac{2x+2}{4x^2+8x+1} dx$.

[illegible]

Question 25 (3 marks)

The function $y = ax^3 + 4x^2 + bx$ has a stationary point at $(1, 6)$.

Determine the values of a and b .

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This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

Questions 11-25 are worth 45 marks in total

Question 26 (6 marks)

A worker invests $\$M$ into an account at the beginning of each month. The account earns 3% p.a. interest, compounded monthly.

- (a) Show that the total amount in the account at the end of the second month, A_2 , is given by $A_2 = M \times (1.0025 + 1.0025^2)$. **2**

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- (b) Show that the amount in the account after n months, A_n , is given by $A_n = 401M(1.0025^n - 1)$. **2**

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Question 26 continues on page 20

Question 26 (continued)

- (c) Find how much is needed to invest at the beginning of each month if the plan is to have \$6 000 at the end of 12 months.

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Question 27 (2 marks)

Find the equation of the tangent to the curve $y = \ln(x^2)$ at the point $(1, 0)$.

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Question 28 (6 marks)

The displacement of a particle is described by the equation $x = 4te^{-t} + 3$, where x is the displacement from the origin in cm and t is the time in seconds.

- (a) Find the particle's initial displacement. 1

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- (b) Find an equation for the particle's velocity. 2

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- (c) Find when the particle is at its maximum distance from the origin. 1

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- (d) Describe what happens to the particle eventually. 2

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Question 29 (4 marks)

A six-sided die is biased as shown in the probability distribution below.

$X = x$	1	2	3	4	5	6
$P(X = x)$	0.1	0.25	0.05	a	0.17	0.13

- (a) Explain why $a = 0.3$. **1**

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- (b) Find the expected value of X . **1**

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- (c) Find the variance of X . **2**

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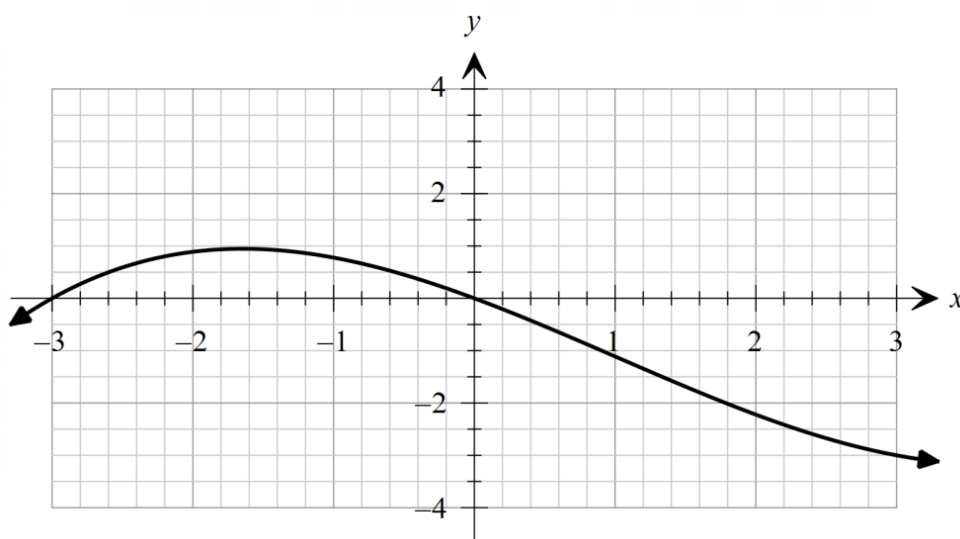
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Question 30 (2 marks)

The graph below shows $y = f(x)$.

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On the same graph, sketch $y = -f(x-1)$.



Question 31 (5 marks)

A probability density function is defined as $f(x) = ke^{-x}$ on the domain $[0, 3]$.

- (a) Show that $k = 1.0524$, correct to four decimal places.

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Question 31 continues on page 24

Question 31 (continued)

- (b) Hence, find the median value of the probability density function, correct to three decimal places.

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Question 32 (2 marks)

Find the value of m given that m , $3m$ and $m^2 + 20$ are consecutive terms of a geometric sequence.

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Question 33 (4 marks)

(a) Find $\frac{d}{dx}(x^3 e^{x^3})$.

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(b) Hence, find the exact value of $\int_0^1 x^2 e^{x^3} (1 + x^3) dx$.

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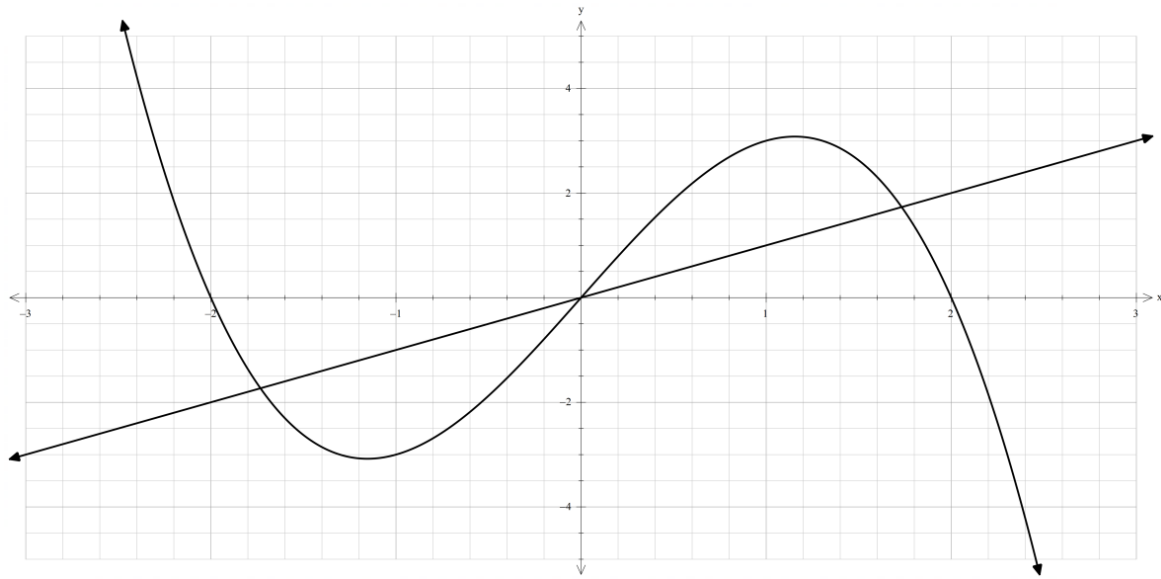
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Question 34 (4 marks)

The functions $y = -x^3 + 4x$ and $y = x$ are sketched below.



- (a) Show that the functions intersect when $x = 0$ and $x = \pm\sqrt{3}$.

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[illegible]

Question 34 continues on page 27

Question 34 (continued)

- (b) Hence find the exact area between the two functions in the first quadrant. **2**

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Question 35 (4 marks)

The function $f(x) = x^2 e^x$ is defined for the domain $[-4, 1]$.

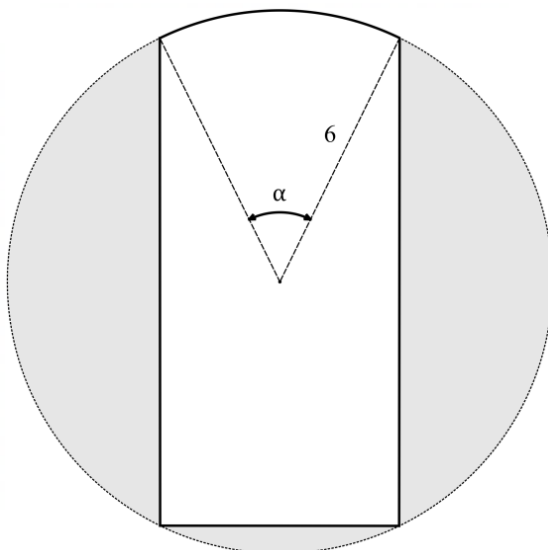
Find the global maximum and minimum values of this function.

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Question 36 (6 marks)

A circular piece of metal of radius 6 cm has three segments removed as shown in the diagram below. The angle α is measured in radians.



- (a) Show that the area of the remaining metal is given by:

$$A = 18(\alpha + 3 \sin \alpha) \text{ cm}^2.$$

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Question 36 continues on page 30

Question 36 (continued)

- (b) Hence, find the angle α , correct to two decimal places, such that the area is a maximum.

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[illegible]

End of Paper

Mathematics Advanced

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface Area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and Series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, \quad r \neq 1$$

$$S = \frac{a}{1-r}, \quad |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2} ab \sin C$$

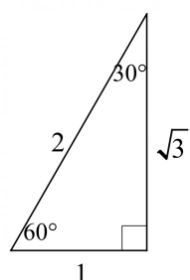
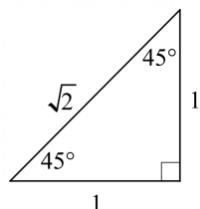
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2} r^2 \theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

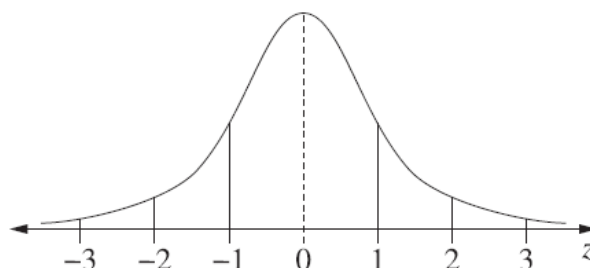
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal Distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^n C_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\text{Var}(X) = np(1-p)$$

Differential Calculus

Function

$$y = f(x)^n$$

$$y = uv$$

$$y = g(u) \text{ where } u = f(x)$$

$$y = \frac{u}{v}$$

$$y = \sin f(x)$$

$$y = \cos f(x)$$

$$y = \tan f(x)$$

$$y = e^{f(x)}$$

$$y = \ln f(x)$$

$$y = a^{f(x)}$$

$$y = \log_a f(x)$$

$$y = \sin^{-1} f(x)$$

$$y = \cos^{-1} f(x)$$

Derivative

$$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \dots + \binom{n}{r}x^{n-r}a^r + \dots + a^n$$

Vectors

$$|\underline{u}| = |\underline{x}\underline{i} + \underline{y}\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda\underline{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta) \\ = re^{i\theta}$$

$$\left[r(\cos\theta + i\sin\theta)\right]^n = r^n(\cos n\theta + i\sin n\theta) \\ = r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Student Name:.....

MATHEMATICS ADVANCED – MULTIPLE-CHOICE ANSWER SHEET

ATTEMPT ALL QUESTIONS

Question	1	A ☐	B ☐	C ☐	D ☐
	2	A ☐	B ☐	C ☐	D ☐
	3	A ☐	B ☐	C ☐	D ☐
	4	A ☐	B ☐	C ☐	D ☐
	5	A ☐	B ☐	C ☐	D ☐
	6	A ☐	B ☐	C ☐	D ☐
	7	A ☐	B ☐	C ☐	D ☐
	8	A ☐	B ☐	C ☐	D ☐
	9	A ☐	B ☐	C ☐	D ☐
	10	A ☐	B ☐	C ☐	D ☐

2020 Higher School Certificate

Mathematics Advanced Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	B
2	A
3	D
4	B
5	C
6	B
7	A
8	D
9	B
10	C

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Section II

Question 11

Criteria	Marks
• Correct derivative using product rule (Does not need to be factorised)	2
• Attempts to use product rule	1

Sample answer

Let $u = x^2$ and $v = \sin x$
 $u' = 2x$ and $v' = \cos x$

$$\frac{d}{dx}(x^2 \sin x) = x^2 \cos x + 2x \sin x$$

Question 12

Criteria	Marks
• Correct integration	2
• Minor error in coefficient or no constant term	1

Sample answer

$$\int \cos 4x \, dx = -\frac{1}{4} \sin 4x + C$$

Question 13

Criteria	Marks
• Correctly uses formula to produce derivative	3
• Makes some progress towards correct derivative after substituting	2
• Substitutes correctly into formula	1

Sample answer

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3x + 3h - x^2 - 3x}{h} \\&= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h} \\&= \lim_{h \rightarrow 0} (2x + 3 + h) \\&= 2x + 3\end{aligned}$$

Question 14a

Criteria	Mark
• Correct answer	1

Sample answer

$$\text{Range} = 55 - 30 = 25$$

Question 14b

Criteria	Mark
• Correct answer	1

Sample answer

$$\text{Median} = 44$$

Question 15a

Criteria	Mark
• Correct answer	1

Sample answer

10th term of arithmetic sequence with $a = 3$ and $d = 2$.

$$\begin{aligned}T_{10} &= 3 + (10 - 1) \times 2 \\ &= 21\end{aligned}$$

Question 15b

Criteria	Marks
• Correct answer	2
• Attempts to find answer using sum of n terms of an arithmetic series	1

Sample answer

$$S_n = 400$$

$$400 = \frac{n}{2}(2 \times 3 + (n - 1) \times 2)$$

$$400 = \frac{n}{2}(2n + 4)$$

$$400 = n^2 + 2n$$

$$\therefore n^2 + 2n - 400 = 0$$

$$\therefore n = \frac{-2 \pm \sqrt{1604}}{2}$$

$$n = 19.024... \text{ or } -42.049...$$

So can make 19 complete rows.

Question 16a

Criteria	Mark
• Correct answer	1

Sample answer

25% of students in Y11 are below 160 cm.

Question 16b

Criteria	Mark
• Correct answer	1

Sample answer

Y11 – 25% above 175 cm which is 35 students.

Y12 – also 35 students above 175 cm but this represents 50% of the year group.

So 70 students in Y12 in total.

Question 17

Criteria	Marks
• Correctly calculates using two applications of trapezoidal rule	3
• Uses two applications of trapezoidal rule, but with minor errors	2
• One application of trapezoidal rule	1

Sample answer

$$\text{Area} \approx \frac{12.5}{2}(15 + 16 + 2 \times 18) + \frac{12.5}{2}(16 + 13 + 2 \times 20) = 850$$

The garden is approximately 850 m².

Question 18a

Criteria	Mark
• Correct answer	1

Sample answer

$$\angle BAC = 270^\circ - 210^\circ = 60^\circ$$

Question 18b

Criteria	Marks
• Correctly calculates distance using cosine rule	2
• Attempts to use cosine rule	1

Sample answer

$$\begin{aligned}x^2 &= 30^2 + 22^2 - 2 \times 30 \times 22 \times \cos(60^\circ) \\&= 724 \\x &= \sqrt{724} \\&= 26.907...\end{aligned}$$

The distance between towns B and C is 27 km.

Question 19a

Criteria	Marks
• Correctly finds the stationary point and it's their nature	3
• Attempts to determine the nature of the point	2
• Finds x value for possible stationary point	1

Sample answer

$$\begin{aligned}y' &= \frac{e^x - xe^x}{e^{2x}} & \text{and} & & y'' &= \frac{(-1)e^x - (1-x)e^x}{e^{2x}} \\&= \frac{1-x}{e^x} & & & &= \frac{x-2}{e^x}\end{aligned}$$

For stationary points, $y' = 0$

$$\begin{aligned}\frac{1-x}{e^x} &= 0 \\ \therefore 1-x &= 0 \\ \therefore x &= 1\end{aligned}$$

When $x = 1$, $y = \frac{1}{e}$ and $y'' = -\frac{1}{e^{-1}} = -e < 0$

Therefore, $\left(1, \frac{1}{e}\right)$ is a maximum turning point.

Question 19b

Criteria	Marks
• Correct graph with correct points and asymptote	4
• Correct shape and points, with incorrect asymptote	3
• Correct shape with incorrect inflection and asymptote	2
• Correctly finds inflection	1

Sample answer

Inflection when $y'' = 0$

$$\frac{x-2}{e^x} = 0$$

$$\therefore x-2 = 0$$

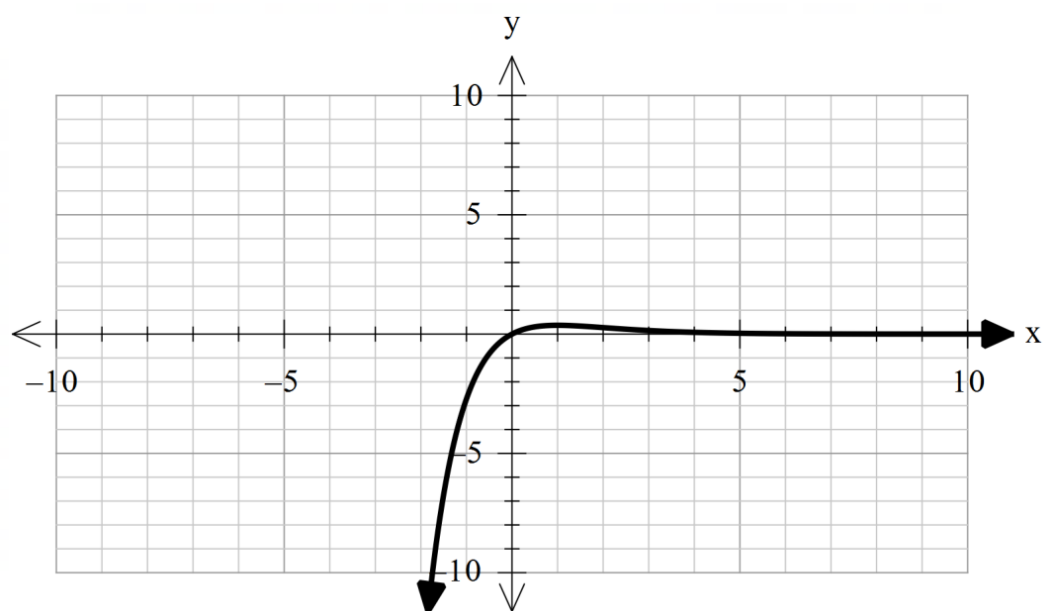
$$\therefore x = 2$$

when $x = 2$, $y = \frac{2}{e^2}$ and $x < 2, y'' < 0$ (concave down) and $x > 2, y'' > 0$ (concave up)

so concavity changes.

$\left(2, \frac{2}{e^2}\right)$ is a point of inflection.

Asymptote: as $x \rightarrow \infty$, $y \rightarrow 0^+$



Question 20a

Criteria	Mark
• Correctly determines the equation of the least-squares regression line	1

Sample answer

$$y = -2.1x + 22.1$$

Question 20b

Criteria	Mark
• Correctly uses their equation to find hours exercising	1

Sample answer

$$15 = -2.1x + 22.1$$

$$2.1x = 7.1$$

$$\therefore x = 3.38\dots$$

Would spend 3.38 hours exercising.

Question 21a

Criteria	Mark
• Correct answer	1

Sample answer

170 cm is a z-score of 2.

So 2.5% of students above 170 cm.

Question 21b

Criteria	Marks
• Correct answer	2
• Finds correct percentage	1

Sample answer

158 cm is a z-score of -1

170 cm is a z-score of 2

From -1 to 2 is 81.5%

Number of students = $200 \times 0.815 = 163$

Question 22

Criteria	Marks
• All possible solutions found	3
• Finds correct solutions for incorrect domain	2
• Finds acute solution	1

Sample answer

$$2\sin(3\theta) = -1 \text{ for } 0 \leq \theta \leq \pi$$

$$\sin(3\theta) = -\frac{1}{2} \text{ for } 0 \leq 3\theta \leq 3\pi$$

$$\text{Acute angle} = \frac{\pi}{6}$$

$$\therefore 3\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore \theta = \frac{7\pi}{18}, \frac{11\pi}{18}$$

Question 23a

Criteria	Mark
• Correct answer	1

Sample answer

Initial population when $t = 0$

$$\begin{aligned}P &= 1000e^0 \\ &= 1000\end{aligned}$$

Question 23b

Criteria	Marks
• Correctly finds the value of k	2
• Substitutes correctly into formula	1

Sample answer

$$\begin{aligned}15000 &= 1000e^{5k} \\ 15 &= e^{5k} \\ 5k &= \ln 15 \\ k &= \frac{\ln 15}{5} \\ k &= 0.5416\dots\end{aligned}$$

Question 23c

Criteria	Marks
• Correctly rounds to the 8 th week	2
• Finds t as decimal	1

Sample answer

$$\begin{aligned}50000 &= 1000e^{0.5461t} \\ 50 &= e^{0.5461t} \\ 0.5461t &= \ln 50 \\ t &= \frac{\ln 50}{0.5461} \\ t &= 7.223\dots\end{aligned}$$

Exceeds 50000 during the 8th week.

Question 24

Criteria	Marks
• Correctly integrates	2
• Identifies numerator as derivative of denominator	1

Sample answer

$$\text{If } f(x) = 4x^2 + 8x + 1$$

$$\text{then } f'(x) = 8x + 8$$

$$\begin{aligned}\int \frac{2x+2}{4x^2+8x+1} dx &= \frac{1}{4} \int \frac{8x+8}{4x^2+8x+1} dx \\ &= \frac{1}{4} \ln|4x^2+8x+1| + C\end{aligned}$$

Question 25

Criteria	Marks
• Correct values for a and b	3
• Establishes two equations to solve simultaneously	2
• Puts $y'=0$ for the point (1,6)	1

Sample answer

$$y = ax^3 + 4x^2 + bx$$

$$y' = 3ax^2 + 8x + b$$

Substitute point:

$$6 = a + 4 + b$$

$$\therefore a + b = 2 \quad - (1)$$

$$y' = 0 \text{ at } (1,6)$$

$$0 = 3a + 8 + b$$

$$\therefore 3a + b = -8 \quad - (2)$$

$$(2) - (1):$$

$$2a = -10$$

$$a = -5$$

$$b = 7$$

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Question 26a

Criteria	Marks
• Correct expression for A_2 from A_1	2
• Correct expression for A_1	1

Sample answer

$$A_1 = M \times 1.0025$$

$$\begin{aligned} A_2 &= (A_1 + M) \times 1.0025 \\ &= (M \times 1.0025 + M) \times 1.0025 \\ &= M \times (1.0025 + 1.0025^2) \end{aligned}$$

Question 26b

Criteria	Marks
• Generates A_n using the sum of a series formula	2
• Expression for A_n with series	1

Sample answer

$$A_3 = M \times (1.0025 + 1.0025^2 + 1.0025^3)$$

$$A_n = M \times (1.0025 + 1.0025^2 + \dots + 1.0025^n)$$

$1.0025 + 1.0025^2 + \dots + 1.0025^n$ is a geometric series with $a = 1.0025$, $r = 1.0025$ and n terms.

$$\begin{aligned} S_n &= \frac{1.0025(1.0025^n - 1)}{1.0025 - 1} \\ &= 401(1.0025^n - 1) \end{aligned}$$

$$A_n = 401M(1.0025^n - 1)$$

Question 26c

Criteria	Marks
• Correct answer	2
• Correct substitution into formula	1

Sample answer

$$A_{12} = 6000$$

$$6000 = 401M(1.0025^{12} - 1)$$

$$M = \frac{6000}{401(1.0025^{12} - 1)}$$

$$\therefore M = \$491.93$$

Question 27

Criteria	Marks
• Correct equation of tangent	2
• Correct derivative	1

Sample answer

$$y' = \frac{2x}{x^2}$$

$$= \frac{2}{x}$$

At the point (1,0)

$$y' = \frac{2}{1}$$

$$\therefore m_T = 2$$

Equation of tangent:

$$y - 0 = 2(x - 1)$$

$$\therefore y = 2x - 2$$

Question 28a

Criteria	Mark
• Correct answer	1

Sample answer

Initial displacement when $t = 0$.

$$x = 4 \times 0 \times e^0 + 3$$

$$\therefore x = 3$$

Initial displacement is 3 cm.

Question 28b

Criteria	Marks
• Correct equation for velocity	2
• Attempts to differentiate displacement equation	1

Sample answer

$$x = 4te^{-t} + 3$$

$$\begin{aligned} v &= \frac{dx}{dt} \\ &= 4e^{-t} - 4te^{-t} \\ &= 4e^{-t}(1-t) \end{aligned}$$

Question 28c

Criteria	Mark
• Correct answer	1

Sample answer

Maximum displacement when $v = 0$, so when $t = 1$.

Question 28d

Criteria	Marks
• Correct answer for x and v	2
• Correct answer for one of x or v	1

Sample answer

As $t \rightarrow \infty$, $e^{-t} \rightarrow 0$ so $x \rightarrow 3^+$ and $v \rightarrow 0$.

Question 29a

Criteria	Mark
• Correct explanation	1

Sample answer

Probability distribution, so $\sum P(X = x) = 1$.

Question 29b

Criteria	Mark
• Correct answer	1

Sample answer

$$\begin{aligned}
 E(x) &= 1 \times 0.1 + 2 \times 0.25 + 3 \times 0.05 + 4 \times 0.3 + 5 \times 0.17 + 6 \times 0.13 \\
 &= 3.58
 \end{aligned}$$

Question 29c

Criteria	Marks
• Correct answer	2
• Some progress	1

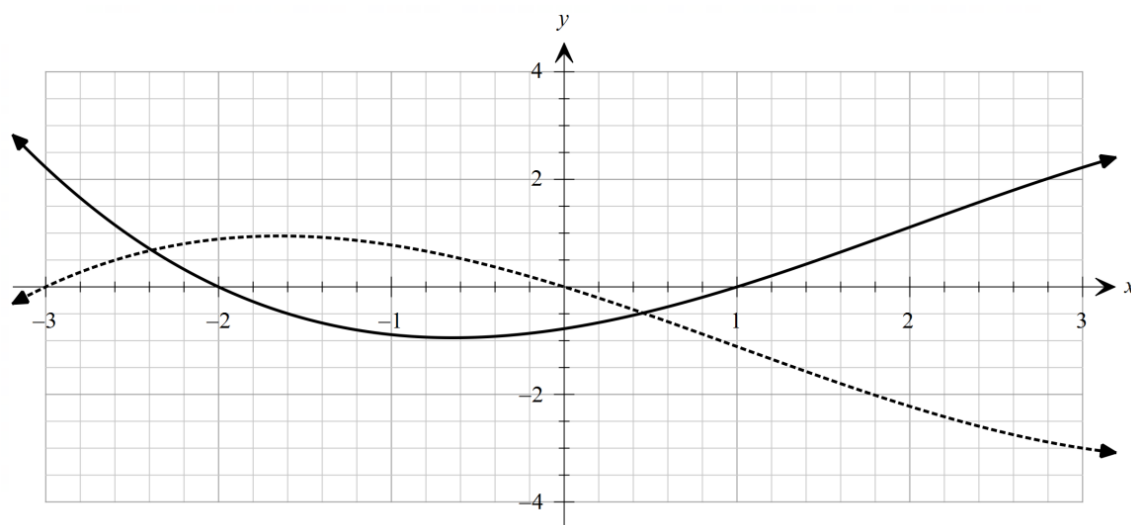
Sample answer

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - \mu^2 \\
 &= 1^2 \times 0.1 + 2^2 \times 0.25 + 3^2 \times 0.05 + 4^2 \times 0.3 + 5^2 \times 0.17 + 6^2 \times 0.13 - 3.58^2 \\
 &= 2.4636
 \end{aligned}$$

Question 30

Criteria	Marks
• Correct graph	2
• Correct reflection or horizontal translation	1

Sample answer



Question 31a

Criteria	Marks
• Correctly shows the value of k	2
• Sets up correct integral	1

Sample answer

$$\int_0^3 k e^{-x} dx = 1$$

$$\left[-k e^{-x} \right]_0^3 = 1$$

$$-k e^{-3} + k e^0 = 1$$

$$k(1 - e^{-3}) = 1$$

$$k = \frac{1}{1 - e^{-3}}$$

$$k = 1.05239\dots$$

$$\therefore k \approx 1.0524$$

Question 31b

Criteria	Marks
• Correct answer	3
• Solves integral correctly	2
• Sets up correct integral	1

Sample answer

$$\int_0^a 1.0524e^{-x} dx = 0.5$$

$$\left[-1.0524e^{-x}\right]_0^a = 0.5$$

$$-1.0524(e^{-a} + 1) = 0.5$$

$$e^{-a} = \frac{0.5 - 1.0524}{-1.0524}$$

$$-a = \ln\left(\frac{0.5 - 1.0524}{-1.0524}\right)$$

$$a = 0.64455\dots$$

$$\therefore a \approx 0.645$$

Question 32

Criteria	Marks
• Solves quadratic for m	2
• Sets up correct ratio equation	1

Sample answer

$$\frac{3m}{m} = \frac{m^2 + 20}{3m}$$

$$9m = m^2 + 20$$

$$m^2 - 9m + 20 = 0$$

$$(m - 4)(m - 5) = 0$$

$$\therefore m = 4, 5$$

Question 33a

Criteria	Marks
• Correct derivative	2
• Attempts to find derivative using product rule	1

Sample answer

$$\begin{aligned}\frac{d}{dx}(x^3 e^{x^3}) &= x^3 \times 3x^2 e^{x^3} + 3x^2 \times e^{x^3} \\ &= 3x^2 e^{x^3} (1 + x^3)\end{aligned}$$

Question 33b

Criteria	Marks
• Correct answer	2
• Uses result from (a) to solve integral	1

Sample answer

$$\begin{aligned}\int_0^1 x^2 e^{x^3} (1 + x^3) dx &= \frac{1}{3} \int_0^1 3x^2 e^{x^3} (1 + x^3) dx \\ &= \frac{1}{3} \left[x^3 e^{x^3} \right]_0^1 \\ &= \frac{1}{3} e\end{aligned}$$

Question 34a

Criteria	Marks
• Solves equation correctly for x	2
• Sets up correct equation	1

Sample answer

$$\begin{aligned}-x^3 + 4x &= x \\ x^3 - 3x &= 0 \\ x(x^2 - 3) &= 0 \\ x = 0 \quad \text{or} \quad x^2 &= 3 \\ \therefore x &= 0, \pm\sqrt{3}\end{aligned}$$

Question 34b

Criteria	Marks
• Solves integral correctly	2
• Correct integral	1

Sample answer

$$\begin{aligned}\int_0^{\sqrt{3}} (-x^3 + 4x - x) dx &= \int_0^{\sqrt{3}} (-x^3 + 3x) dx \\&= \left[-\frac{x^4}{4} + \frac{3x^2}{2} \right]_0^{\sqrt{3}} \\&= \left(-\frac{(\sqrt{3})^4}{4} + \frac{3(\sqrt{3})^2}{2} \right) - 0 \\&= -\frac{9}{4} + \frac{9}{2} \\&= \frac{9}{4}\end{aligned}$$

Area is $\frac{9}{4}$ units²

Question 35

Criteria	Marks
• Finds correct maximum and minimums	4
• Finds max/min turning points and checks endpoints of domain	3
• Finds max/min turning points only	2
• Finds x values of stationary points only	1

Sample answer

$$\begin{aligned}f'(x) &= 2xe^x + x^2e^x \\ &= xe^x(2+x)\end{aligned}$$

$$\begin{aligned}f''(x) &= 2e^x + 2xe^x + 2xe^x + x^2e^x \\ &= e^x(2+4x+x^2)\end{aligned}$$

$$f'(x) = 0 \text{ for min/max}$$

$$\begin{aligned}xe^x(2+x) &= 0 \\ \therefore x &= 0, -2\end{aligned}$$

when $x = -2$, $f(-2) = 4e^{-2} \approx 0.541$ and $f''(x) < 0$ so this is a max

when $x = 0$, $f(0) = 0$ and $f''(x) > 0$ so this is a min

Check endpoints of domain:

$$\text{when } x = -4, f(-4) = 16e^{-4} \approx 0.293$$

$$\text{when } x = 1, f(1) = e \approx 2.718$$

Global max is 2.718

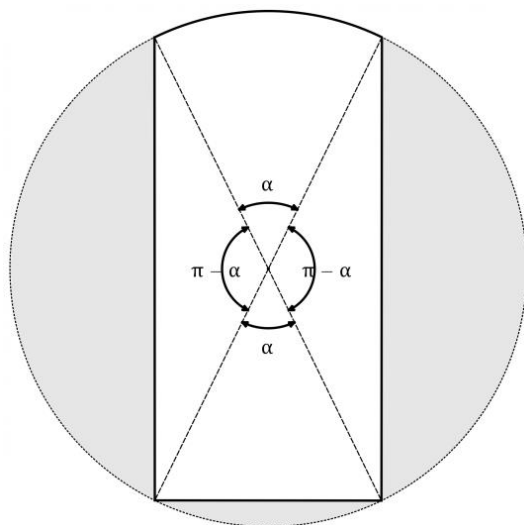
Global min is 0

Question 36a

Criteria	Marks
• Shows correct expression	3
• Find area of two equal triangles using $\sin(\pi - \alpha) = \sin \alpha$	2
• Find area of sector only	1

Sample answer

The shape can be split into a sector and three triangles.



Area sector:

$$A = \frac{1}{2} \times 6^2 \times \alpha = 18\alpha$$

Area of triangle opposite sector:

$$A = \frac{1}{2} \times 6^2 \times \sin \alpha = 18 \sin \alpha$$

Area of other two equal triangles:

$$A = 2 \times \frac{1}{2} \times 6^2 \times \sin(\pi - \alpha) = 2 \times 18 \times \sin \alpha$$

Total Area:

$$\begin{aligned} A &= 18\alpha + 18 \sin \alpha + 2 \times 18 \sin \alpha \\ &= 18(\alpha + 3 \sin \alpha) \end{aligned}$$

Question 36b

Criteria	Marks
• Shows that it is a maximum	3
• Solves $A' = 0$ correctly	2
• Puts $A' = 0$	1

Sample answer

$A' = 0$ for max/min

$$A' = 18(1 + 3 \cos \alpha)$$

$$18(1 + 3 \cos \alpha) = 0$$

$$1 + 3 \cos \alpha = 0$$

$$\cos \alpha = -\frac{1}{3}$$

$$\therefore \alpha \approx 1.91$$

$$A'' = 18(-3 \sin \alpha)$$

$$= -54 \sin \alpha$$

when $\alpha = 1.91$, $A'' = -54 \sin(1.91) = -50.923... < 0$

So maximum area occurs when $\alpha = 1.91$

**2020 Higher School Certificate
Trial Examination
Mathematics
Mapping Grid**

Section I

Question	Marks	Content	Syllabus Outcomes
1	1	Differentiation of power and exponential	MA-C1.4,MA-C2.1
2	1	Simplifying algebraic expressions	MA-F1.1
3	1	Amplitude of sine curve	MA-T3
4	1	Mean and standard deviation	MA-S2.1*
5	1	Expected value of discrete distribution	MA-S1
6	1	Definite integral of powers	MA-C4.2
7	1	Right-angled trigonometry	MA-T1.1*
8	1	Logarithmic laws	MA-E1.2
9	1	Transformation of graphs	MA-F2
10	1	Derivative from graph	MA-C3.1

Section II

Question	Marks	Content	Syllabus Outcomes
11	2	Product rule	MA-C2
12	2	Integral of trigonometric function	MA-C4.1
13	3	First principles differentiation	MA-C1.3
14	2	Range and Median	MA-S2.1*
15	3	Arithmetic sequence	MA-M1.2
16	2	Box plot	MA-S2.1*
17	3	Trapezoidal rule	MA-C4.2*
18	3	Non-right-angled trigonometry	MA-T1.1*
19	7	Graphing using calculus	MA-C3.1, MA-F2, MA-E1.4
20	2	Bivariate data	MA-S2.2*
21	3	Normal distribution	MA-S3.2*
22	3	Trigonometric equation	MA-T2
23	5	Exponential growth	MA-E1.4
24	2	Integration	MA-C4.1
25	3	Stationary points	MA-C3.1
26	6	Financial application of series	MA-M1.4
27	2	Equation of tangent	MA-C1.4, MA-C2
28	6	Displacement of particle	MA-C1.4, MA-C2
29	4	Discrete probability distribution	MA-S1.2
30	2	Transformation of a graph	MA-F2
31	5	Continuous probability distribution	MA-S3.1
32	2	Geometric sequence	MA-M1.3
33	4	Differentiate and Integrate	MA-C2.2, MA-C4.2
34	4	Area between curves	MA-C4.2, MA-F1.3
35	4	Global max/min	MA-C3.2
36	6	Optimisation problem	MA-C3.2, MA-T2